System Update - Known Issues & Troubleshooting

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Team Owner	Sysuppers
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    - How do I fix corrupt RPM database?

Known Issues

Issue Description	Affects Version	Ticket #	Workaround	Fixed in	Comments

The Patch bundle import task fails to install the patch hook.
 	
How to diagnose it:
 	
1. During the importation the UI will only show the message `Starting import for file id <Some ID>`
2. If review the logs of backend with the command `tail -f /var/log/em7/patch_manager.log` the output will show only a similar logs as are next:
 	
`2024-07-03 15:18:19,096 - root - INFO - Installing patch hook`
`2024-07-03 15:19:50,379 - root - INFO - Installing patch hook`
`2024-07-03 15:21:21,586 - root - INFO - Installing patch hook`
`2024-07-03 15:22:52,683 - root - INFO - Installing patch hook`
`2024-07-03 15:24:23,697 - root - INFO - Installing patch hook`
`2024-07-03 15:25:54,781 - root - INFO - Installing patch hook`
`2024-07-03 15:27:25,877 - root - INFO - Installing patch hook`
`2024-07-03 15:28:58,875 - root - INFO - Installing patch hook`
`2024-07-03 15:30:32,334 - root - INFO - Installing patch hook`
`2024-07-03 15:32:04,541 - root - INFO - Installing patch hook`
`2024-07-03 15:33:35,665 - root - INFO - Installing patch hook`
	12.2.0	EM-63105	1. Review if the folder `.rpmdb` there is in the environment with the next command `ls -lha /var/lib/em7/update/patch_hook/.rpmdb`
			2. Create the folder and change the owner and group with the next commands:
			
			`sudo mkdir -p /var/lib/em7/update/patch_hook/.rpmdb`
			`sudo chown -R root:s-em7-core /var/lib/em7/update/patch_hook/`
			
			3. Restart the service `siloupdate-manager`
			
			`sudo systemctl restart siloupdate-manager`
				12.2.1	

deploy fails on collector with post update error on 10.1.x+ and 10.2.x+	10.1.x - 10.2.x	EM-44278: Post_update Failing to Load Docker Images on a collector CLOSED	Postupdate failing to load Docker images on a collector Docker Postupdate and Patch Failures on 10.2 and 11.1 lines	11.1.0 EM-44335: Make reload and restart containers script more robust CLOSED	
module_upgrade_mariadb script fails on appliances configured with non-default DB password	10.1.x	EM-38234: (addressed with EM-38	**If the script is already run and it failed to execute mysql_upgrade** 1. Confirm if the mysql_upgrade command failed by running the following command on failed appliance	10.2.0 EM-38378: Refact	

| | | 378) CISCO CMS - Upgrade mariadb update script to retrieve passwd (incase of non-default) from silo.conf CLOSED<br><br>EM-3922 0: Upgrading Lab SL1 to 10.1.1 issue updating MariaDB CLOSED | ```
Siva-MAC ~$ grep -i 'mysql_upgrade' -A5 /data/logs/upgrade_mariadb.log
[2020-09-01 17:45:43] ## Executing [mysql_upgrade -u root -pem7admin] ##
[2020-09-01 17:45:43] Version check failed. Got the following error when calling the 'mysql' command line client
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: YES)
FATAL ERROR: Upgrade failed
[2020-09-01 17:45:43] Command completed with exit code 1
[2020-09-01 17:45:43]
Siva-MAC ~$
```<br><br>2. Run the following command manually on the Primary DB if mysql_upgrade failed on Primary DB and this error is seen:<br><br>```<br>mysql_upgrade -u root -p<custom_password> (if required run with -force option)<br>systemctl stop em7 (pls take care of putting the nodes into maintenance mode on HA/DR clusters)<br>systemctl start em7<br>```<br><br>3. Update /opt/em7/bin/.upgrade_mariadb with following changes **on all appliances**:<br>-search for line starting with exec_cmd "mysql_upgrade -u root -pem7admin"<br>-Replace -pem7admin as -p<custom_password><br>4. Re-run module_upgrade -v 10.4.12 -m <mid(s)><br><br>**How to prevent running into this issue(manual workaround):**<br><br>```<br>Before running module_upgrade_mariadb, Run through Steps-3 and Step-4 above on all appliances including DB(s)<br>```<br><br>or mariadb upgrade script (SL1 10.2.0) CLOSED | | |
| **silo.log owned by root instead of s-em7-core**

Sometime system_status script ends up changing the ownership of /var/log/em7/silo.log which can cause all patching operations(Import, Stage & Deploy) to be stuck/fail | 8.14.0 | EM-36128 | Resolution:

Run the following on affected appliances

```
chown s-em7-core:s-em7-core /var/log/em7/silo.log
```

Deploy stuck clearing cache

monitor_deploy script show deploy process gets stuck with child process | 10.1.0 | |
| | 8.14.0+ | EM-35167 | On failed appliance:

1. from monitor_deploy script find the pid for the stuck | Not Available | pa |
```

trying to clear cache - find /dev/shm -type d -name silo_php_cache*			process(mostly find /dev/shm -type d -name silo_php_cache*) and kill it with
			`kill -9 <pid>`
			2. Wait for 2 minutes for deploy process (/bin/bash /usr/libexec/siloupdate/deploy --enablerepo=em7,em7-os --force-postupdate) to complete. If not kill the same
			`kill -9 <pid>`
			3. Update
			`deploy_code=0`
			in
			`/var/lib/em7/.deploy_stamp`
			4. [If HA node], execute
			`crm node ready`
			5. Execute
			`rm -f /tmp/.proc_mgr_pause`
			6. Execute monitor_deploy -n1 and confirm proc_mgr is unpaused and deploy-module status Completed		

Staging gets stuck Staging get stuck in Scheduled state (and not moving into In-Progress) for more than 10 minutes	8.12.0+	EM-31690	1. Set the status to Canceled state for the staging schedule that has incorrect patch_file_id *silo_mysql -e "update siloupdate.staging_schedule ss left join master_logs.system_patches sp on sp.patch_file_id=ss.patch_file_id set state=5 where sp.patch_file_id IS NULL;"* 2. Initiate staging again from UI	8.12.1.2	
pigz issue Deploy timed out and running monitor_deploy on the appliance show deploy gets stuck with pigz in process tree output for longer duration	8.12.0	EM-29028 EM-30428	How to avoid issue(before deploying 8.12.0/8.12.0.1): https://support.sciencelogic.com/s/article/1422 To Recover from issue: Steps to resolve yum stuck with running pigz process	8.12.0.2	
Insufficient disk space Staging often fails due to insufficient disk space(less than 4GB) on `/var`	8.12.0	EM-30506	Adding more disk space to Azure Instance	8.12.0.2	

partition					
RPM DB corruption Happens often on CDBs with phonehome collectors	8.12.0	EM-36126	See section about "How do I fix RPM DB corruption?" on this page	8.12.3.1	
Subsequent attempt to deploy causes failure citing package upgrade issue Happens when deployment times-out for one or more appliance and a subsequent deployment schedule is created immediately	8.12.0	EM-33789	See section about "How do I fix a failed deployment?"	10.1.0	
1. Primary DB appliance went into hung state once the reboot command was triggered. 2. Once primary DB appliance was manually rebooted through Vcenter, the primary DB would not allow the team to promote it.		**SF case - 00230821**	1. Reboot DB appliance through Vcenter 2. Remove individual instance from maintenance mode.		This issue is regarding a Equinix stack hosted in DC3 and you can refer to the ticket **00230821** for more information

Frequently Asked Questions

[System Update(YUM Offline): FAQ]

[YUM Offline: Support Q&A]

Logs needed for Escalation

Before escalating an issue, please review the following **escalation summary page** if similar issue is already captured

[SysUppers Escalations and Outages]

- Output of system update troubleshooting/diagnostics tool
  `sudo sysuptb all`
- Following Logs and table dumps

```
mysqldump -uroot -pem7admin siloupdate  > /tmp/sysup-escalation-dump.sql
mysqldump -uroot -pem7admin  master_platform appliances >> /tmp/sysup-escalation-dump.sql
mysqldump -uroot -pem7admin  master_logs system_patches >> /tmp/sysup-escalation-dump.sql
mysqldump -uroot -pem7admin  master system_settings_licenses >> /tmp/sysup-escalation-dump.sql
tar -zcvf sysup-escalation-logs.tgz /data/logs/{patch*.log*,silo.log*} /var/log/yum.log /tmp/sysup-escalation-dump.sql
```

- Snapshot of `monitor_stage` or `monitor_deploy` output
- Snapshot of Updates UI Page(and stage/deploy log windows) and Appliances Page
- Output of `system_status` script on the primary DB as well as the affected appliance

Troubleshooting

General Checks

- Check if essential services are running on the primary DB, if not start them
  - Task manager (em7.service)
  - Spool Server (siloupdate-spool.service)
  - Pkgserver (siloupdate-pkgserver.service)
  - Collector Task Manager (em7_ctaskman.service)
  - High Frequency Pull (em7_hfpulld.service)

- Check if the ownership of `/var/log/em7/silo.log` file is correct and owned by `s-em7-core:s-em7-core` user and group

```
ls -l /var/log/em7/silo.log
chown s-em7-core:s-em7-core /var/log/em7/silo.log
```

How do I resolve module-cmd not responding leading to patching operations timing out?

Patching operations depends on “module-cmd”s to trigger commands on remote appliances.
A lot of backend processes are involved in order for the command spooled from the primary DB to be delivered to the remote appliances, to be executed locally and bring back the results of the command back the DB.

In order to test, execute the following cmd from CDB:

`timeout -sHUP 2m siloupdate module-cmd <mid> 'hostname'`

on successful completion, this would return the hostname of the <mid> appliance. if it didn’t complete within 2 minutes, then check the following:

On CDB:

- is `em7_hfpulld` process up and running?
- is `em7_ctaskman` systemd service up and running?
- is the remote appliance online(For collectors: `silo_mysql -h <remote_IP> -P 7707 -e "select @@hostname"`)?

On remote appliance:

- is `em7` systemd service up and running?
  - if inactive/dead or if it restarts every one minute, check `/var/log/sl1/em7.log`
- is `proc_mgr` paused?
- is appliance licensed?
- any anomalies with `system_status` script?
  - appliance not being licensed
  - backlogged async processes for admin toolbox or other spooled entries
  - high cpu load/vCPU contention on hosted platforms

How do I check the installed version or upgrade patch hook/siloupdate package?

- To check the installed version of `em7-patch-hook` and `siloupdate` package

```
ls -l /var/lib/em7/update/patch_hook/.rpmdb && rpm -qa em7-patch-hook --dbpath /var/lib/em7/update/patch_hook/.rpmdb/;rpm -q siloupdate;
```

- To check if there are any errors installing either of the two packages

```
grep -A10 -B10 -i 'hook.*error' /data/logs/patch_manager.log
```

- To manually upgrade `em7-patch-hook` package

  If needed download the latest/target version of the `em7-patch-hook` package

```
#Search filestore for list of available packages
pcli filestore list 'em7-patch-hook*'

#Download the package
pcli filestore download 'em7-patch-hook-<full-name>'

#Upgrade the package
rpm -Uvh --nodeps --dbpath /var/lib/em7/update/patch_hook/.rpmdb </Path/To/Package>
```

- To manually upgrade `siloupdate` package

```
#Search filestore for list of available packages
pcli filestore list 'siloupdate*'

```

```
#Download the package
pcli filestore download 'siloupdate-<full-name>'

#Upgrade the package
yum upgrade --nogpgcheck --disablerepo=* -y </Path/To/Package>
```

Import

How do I resolve if a patch import ended up with Failed or Missing Base error?

Refer [How to resolve Import Failed or Missing Base error]

Staging

How do I resolve Staging failure due to "Failed to cache packages" error?

- Identify the error/reason why Cache packages failed by running following command

```
[root@10-64-171-200-CDB ~]# grep -B3 'Failed to cache packages' /data/logs/patch_manager.log| tail -3
"2020-05-18 13:52:40,200","DEBUG","20154","161","patch_stager","344","sid=11, 2020/05/18 13:52:40 symlink /var/lib/siloupdate/cache/788
/var/lib/siloupdate/packages/el7/em7-os/pixman-0.34.0-1.el7.x86_64.rpm: no space left on device"
"2020-05-18 13:52:40,204","DEBUG","20154","161","patch_stager","349","sid=11, Cache action completed with exit code 1"
"2020-05-18 13:52:40,206","DEBUG","20154","161","db_logger","56","sid=11, Failed to cache packages"
[root@10-64-171-200-CDB ~]#
```

- If the error is about insufficient disk space, clean up the appropriate partition(by default either /tmp or /var/lib whichever has larger "available disk" space)

> Alternatively another new partition having larger available disk space can be added
>
> pcli set-patcher-param repo_cache_dir </path/to>

For more info refer [Patcher CLI]

Adding new directory for caching packages

- In the above example, Cache package failed due to insufficient disk space on /var/lib partition.

- Based on df -h, / partition seem to have enough available space(~3GB)

- pcli list-patcher-param lists the currently available directories to cache packages(the highest available disk space will be selected)

```
[root@10-64-171-200-CDB ~]# pcli list-patcher-param
+-------------------------+----------+-----------------------------------------------------------------------------------------+
| Parameter | Value | Description |
+=========================+==========+=========================================================================================+
| deploy_wait_time | 3600 | Timeout in seconds for deployment to complete for one appliance |
+-------------------------+----------+-----------------------------------------------------------------------------------------+
| patch_hook_task_timeout | 300 | Timeout in seconds for patch hook to be installed for one appliance |
+-------------------------+----------+-----------------------------------------------------------------------------------------+
| pool_size | 25 | Number of simultaneous processes to spawn for staging and deployment |
+-------------------------+----------+-----------------------------------------------------------------------------------------+
| deploy_retries | 0 | Number of times to retry deployment after the first attempt fails |
+-------------------------+----------+-----------------------------------------------------------------------------------------+
| staging_retries | 0 | Number of times to retry staging after the first attempt fails |
+-------------------------+----------+-----------------------------------------------------------------------------------------+
| staging_wait_time | 4000 | Timeout in seconds for staging to complete for one appliance |
+-------------------------+----------+-----------------------------------------------------------------------------------------+
| repo_cache_dir | /tmp | Directory to use for caching packages for staging. It is a multi valued parameter |
+-------------------------+----------+-----------------------------------------------------------------------------------------+
| repo_cache_dir | /var/lib | Directory to use for caching packages for staging. It is a multi valued parameter |
+-------------------------+----------+-----------------------------------------------------------------------------------------+
| staging_force_upload | 0 | Whether to force upload packages during staging |
+-------------------------+----------+-----------------------------------------------------------------------------------------+
```

- Using pcli set-patcher-param command, / partition is added another directory to cache packages

```
[root@10-64-171-200-CDB ~]# pcli set-patcher-param repo_cache_dir /
```

```
[root@10-64-171-200-CDB ~]# pcli list-patcher-param
+-------------------------+----------+-----------------------------------------------------------------------------------------+
```

```
| Parameter | Value | Description |
+=========================+==========+=============================================
=====================================+
| deploy_wait_time | 3600 | Timeout in seconds for deployment to complete for one
appliance |
+-------------------------+----------+---------------------------------------------
-------------------------------------+
| patch_hook_task_timeout | 300 | Timeout in seconds for patch hook to be installed
for one appliance |
+-------------------------+----------+---------------------------------------------
-------------------------------------+
| pool_size | 25 | Number of simultaneous processes to spawn for staging and
deployment |
+-------------------------+----------+---------------------------------------------
-------------------------------------+
| deploy_retries | 0 | Number of times to retry deployment after the first attempt
fails |
+-------------------------+----------+---------------------------------------------
-------------------------------------+
| staging_retries | 0 | Number of times to retry staging after the first attempt
fails |
+-------------------------+----------+---------------------------------------------
-------------------------------------+
| staging_wait_time | 4000 | Timeout in seconds for staging to complete for one
appliance |
+-------------------------+----------+---------------------------------------------
-------------------------------------+
| repo_cache_dir | /tmp | Directory to use for caching packages for staging. It is a
multi valued parameter |
+-------------------------+----------+---------------------------------------------
-------------------------------------+
| repo_cache_dir | /var/lib | Directory to use for caching packages for staging. It
is a multi valued parameter |
+-------------------------+----------+---------------------------------------------
-------------------------------------+
| staging_force_upload | 0 | Whether to force upload packages during staging |
+-------------------------+----------+---------------------------------------------
-------------------------------------+
| repo_cache_dir | / | None |
+-------------------------+----------+---------------------------------------------
-------------------------------------+
[root@10-64-171-200-CDB ~]#
```

```
```

How do I resolve staging failure due to “Staging will not be attempted for appliance <mid>. There is no platform information available”?

Please capture logs with debug logging enabled for SiloUpdate.

(All steps in this section should be applied to the CDB unless it is stated otherwise)

If there are log messages that indicate a version query timeout

This is the most common type of issue and is indicated by the following message:

```
DEBUG - Version query result from appliance <mid>: (<mid>, TimeoutError(), None)
```

This could be a symptom of “module_version_query_timeout” being set to a value that is too low.

To troubleshoot, start by entering a mysql prompt with `silo_mysql` and running the following query:

```
select * from master.system_settings_patcher where param="module_version_query_timeout" \G
```

If there are **no results**: insert a new row by executing the following (which sets version query timeout to 5 seconds):

```
insert into master.system_settings_patcher (param, value) values ("module_version_query_timeout", "5000");
```

If there is a **single result** and the value of `value` is less than a few seconds or so (it will be given in milliseconds): consider raising the timeout by executing the following:

```
update master.system_settings_patcher set value="5000" where param="module_version_query_timeout";
```

(In cases of multiple results or non-numeric values of `value` then there has been a misconfiguration.)

If the problem persists after increasing “module_version_query_timeout”, it may be a symptom of a range of problems on the CU side. Please document steps taken so far and SysUppers may assist further.

If there are other errors/exceptions

This may be a symptom of a range of problems with the stack, please bring to the attention of SysUppers.

How do I increase the default Staging time out?

This is indicated by debug log messages at the end of staging that look like the following:

```
Patch staging timed out on module <mid>
```

It can also be indicated while staging is running with log messages that reference `asyncio.CancelledError` or `staging_wait_time`.

From the primary DB or AIO, execute the following command.

8.14.0 release onwards

```
sudo pcli set-patcher-param staging_wait_time <timeout_in_seconds>
```

pcli, a new command line utility for patcher is available from 8.14.0+

Increasing stage wait time to 2 hours

```
[root@ip-10-152-4-100 ~]# pcli set-patcher-param staging_wait_time 7200
[root@ip-10-152-4-100 ~]#
[root@ip-10-152-4-100 ~]# pcli list-patcher-param
+------------------------+----------+-----------------------------------------------------------------------------------------+
|        Parameter       |  Value   |                                       Description                                       |
```

```
+=========================+==========+=============================================================================+
| deploy_wait_time        | 3600     | Timeout in seconds for deployment to complete for one appliance             |
+-------------------------+----------+-----------------------------------------------------------------------------+
| patch_hook_task_timeout | 300      | Timeout in seconds for patch hook to be installed for one appliance         |
+-------------------------+----------+-----------------------------------------------------------------------------+
| pool_size               | 25       | Number of simultaneous processes to spawn for staging and deployment        |
+-------------------------+----------+-----------------------------------------------------------------------------+
| deploy_retries          | 0        | Number of times to retry deployment after the first attempt fails           |
+-------------------------+----------+-----------------------------------------------------------------------------+
| staging_retries         | 0        | Number of times to retry staging after the first attempt fails              |
+-------------------------+----------+-----------------------------------------------------------------------------+
| staging_wait_time       | 7200     | Timeout in seconds for staging to complete for one appliance                |
+-------------------------+----------+-----------------------------------------------------------------------------+
| repo_cache_dir          | /tmp     | Directory to use for caching packages for staging. It is a multi valued parameter |
+-------------------------+----------+-----------------------------------------------------------------------------+
| repo_cache_dir          | /var/lib | Directory to use for caching packages for staging. It is a multi valued parameter |
+-------------------------+----------+-----------------------------------------------------------------------------+
[root@ip-10-152-4-100 ~]#
```

For Pre-8.14.0 releases

master.system_settings_patcher: contains [user-configurable parameters for patcher](#).

```
[root@dw-sm-aio-patch-10-2-12-111 em7admin]# silo_mysql -e "select * from master.system_settings_patcher"
+----------+-------------------------+----------+---------------------+-------------------------------------------------------------------------
```

siloupdate.staging_schedule: Contains staging schedules and attempts information

```
MariaDB [siloupdate]> select * from siloupdate.staging_schedule;
+------------+---------------+------------------+------------------+-------+---------------------+
| staging_id | patch_file_id | staged_appliances | failed_appliances | state | edit_date           |
+------------+---------------+------------------+------------------+-------+---------------------+
|          1 |             0 |                0 |                0 |     0 | 2019-06-04 08:37:42 |
+------------+---------------+------------------+------------------+-------+---------------------+
1 row in set (0.00 sec)

MariaDB [siloupdate]>
```

siloupdate.staging_logs: Contains deployment log information

```
MariaDB [siloupdate]> select * from siloupdate.staging_logs;
+--------+------------+-----------------------------------+-----+---------------------+
| log_id | staging_id | msg                               | mid | log_stamp           |
+--------+------------+-----------------------------------+-----+---------------------+
|      1 |          1 | Found 4 appliance(s) for deployment |   1 | 2019-06-04 08:37:42 |
+--------+------------+-----------------------------------+-----+---------------------+
1 row in set (0.00 sec)

MariaDB [siloupdate]>
```

For whatever reason, if staging status for one or more appliance need to be reset, following command can be executed

```
delete from siloupdate.appliance_staging where mid=<mid> and patch_file_id=<patch_file_id>
```

`mid => Can be retrieved from System -> Settings -> Appliances Page`
`patch_file_id => Can be retrieved from master_logs.system_patches table or hover the mouse over Stage or Deploy Icon in System -> Tools -> Updates Page and URL link at browser's bottom left status bar will have this info`

Along with staging status, If staging logs also need to be cleared, following command can be executed

```
delete from siloupdate.staging_logs where mid=<mid> and staging_id=<staging_id>
```

`mid => Can be retrieved from System -> Settings -> Appliances Page`
`staging_id => Can be retrieved from siloupdate.staging_schedule`

Pre-Upgrade

[Pre-upgrade troubleshooting]

Deployment

How do I increase the default deploy time out?

From the primary DB or AIO, execute the following command

8.14.0 release onwards

```
sudo pcli set-patcher-param deploy_wait_time <timeout_in_seconds>
```

pcli, a new command line utility for patcher is available from 8.14.0+

Increasing deploy wait time to 2 hours

```
[root@ip-10-152-4-100 ~]# pcli set-patcher-param deploy_wait_time 7200
[root@ip-10-152-4-100 ~]#
[root@ip-10-152-4-100 ~]# pcli list-patcher-param
+-------------------------+----------+-------------------------------------------------------------------------------------+
|        Parameter        |  Value   |                                     Description                                     |
+=========================+==========+=====================================================================================+
| deploy_wait_time        | 7200     | Timeout in seconds for deployment to complete for one appliance                     |
+-------------------------+----------+-------------------------------------------------------------------------------------+
| patch_hook_task_timeout | 300      | Timeout in seconds for patch hook to be installed for one appliance                 |
+-------------------------+----------+-------------------------------------------------------------------------------------+
| pool_size               | 25       | Number of simultaneous processes to spawn for staging and deployment                |
+-------------------------+----------+-------------------------------------------------------------------------------------+
| deploy_retries          | 0        | Number of times to retry deployment after the first attempt fails                   |
+-------------------------+----------+-------------------------------------------------------------------------------------+
| staging_retries         | 0        | Number of times to retry staging after the first attempt fails                      |
+-------------------------+----------+-------------------------------------------------------------------------------------+
| staging_wait_time       | 1800     | Timeout in seconds for staging to complete for one appliance                        |
+-------------------------+----------+-------------------------------------------------------------------------------------+
| repo_cache_dir          | /tmp     | Directory to use for caching packages for staging. It is a multi valued parameter |
+-------------------------+----------+-------------------------------------------------------------------------------------+
| repo_cache_dir          | /var/lib | Directory to use for caching packages for staging. It is a multi valued parameter |
+-------------------------+----------+-------------------------------------------------------------------------------------+
[root@ip-10-152-4-100 ~]#
```

For Pre-8.14.0 releases

master.system_settings_patcher: contains user-configurable parameters for patcher.

```
[root@dw-sm-aio-patch-10-2-12-111 em7admin]# silo_mysql -e "select * from master.system_settings_patcher"
+----------+-------------------------+----------+---------------------+-------------------------------------------------------------------------------------------------------------------------------------+
| param_id | param                   | value    | edit_date           | description                                                                                                                         |
+----------+-------------------------+----------+---------------------+-------------------------------------------------------------------------------------------------------------------------------------+
|        1 | deploy_wait_time        | 3600     | 2019-08-22 18:55:06 | Timeout in seconds for deployment to complete for one appliance                                                                     |
|        2 | patch_hook_task_timeout | 300      | 2019-08-22 18:55:06 | Timeout in seconds for patch hook to be installed for one appliance                                                                 |
|        3 | pool_size               | 25       | 2019-08-22 18:55:06 | Number of simultaneous processes to spawn for staging and deployment                                                                |
|        4 | deploy_retries          | 0        | 2019-08-22 18:55:06 | Number of times to retry deployment after the first attempt fails                                                                   |
|        5 | staging_retries         | 0        | 2019-08-22 18:55:06 | Number of times to retry staging after the first attempt fails                                                                      |
|        6 | staging_wait_time       | 1800     | 2019-08-22 18:55:06 | Timeout in seconds for staging to complete for one appliance                                                                        |
|        7 | repo_cache_dir          | /tmp     | 2019-05-15 17:00:14 | Directory to use for caching packages for staging. It is a multi valued parameter |
|        8 | repo_cache_dir          | /var/lib | 2019-05-15 16:59:57 | Directory to use for caching packages for staging. It is a multi valued parameter |
+----------+-------------------------+----------+---------------------+-------------------------------------------------------------------------------------------------------------------------------------+
[root@dw-sm-aio-patch-10-2-12-111 em7admin]#
```

Command to update default deploy timeout

```
silo_mysql -e "UPDATE master.system_settings_patcher SET value=<TIME_IN_SECONDS> WHERE param='deploy_wait_time'"
```

How do I reset deployment status for an appliance(in order to re-deploy)

siloupdate.appliance_deployment: Contains deployment status information for each appliances

```
MariaDB [(none)]> select * from siloupdate.appliance_deployment;
+-----------+---------------+-----+-------+---------------------+-----------+
| deploy_id | patch_file_id | mid | state | edit_date           | exit_code |
+-----------+---------------+-----+-------+---------------------+-----------+
|         1 |             0 |   1 |     0 | 2019-05-06 18:09:43 |         0 |
+-----------+---------------+-----+-------+---------------------+-----------+
1 row in set (0.00 sec)

MariaDB [(none)]>
```

siloupdate.deployment_schedule: Contains deployment schedules and attempts information

```
MariaDB [(none)]> select * from siloupdate.deployment_schedule;
+-----------+---------------+-------+---------------------+
| deploy_id | patch_file_id | state | edit_date           |
+-----------+---------------+-------+---------------------+
|         1 |             0 |     0 | 2019-05-06 18:09:43 |
+-----------+---------------+-------+---------------------+
1 row in set (0.00 sec)

MariaDB [(none)]>
```

siloupdate.deployment_logs: Contains deployment log information

```
MariaDB [siloupdate]> select * from deployment_logs;
+--------+-----------+-------------------------------------------------------------------------------------------------------------+---
--+---------------------+
| log_id | deploy_id | msg                                                                                                         |
mid | log_stamp           |
+--------+-----------+-------------------------------------------------------------------------------------------------------------+---
--+---------------------+
|      8 |         3 | Found 4 appliance(s) for deployment                                                                         |
1 | 2019-04-25 03:08:45 |
|      9 |         3 | Beginning deployment to module 5                                                                            |
5 | 2019-04-25 03:08:51 |
+--------+-----------+-------------------------------------------------------------------------------------------------------------+---
--+---------------------+
2 rows in set (0.01 sec)

MariaDB [siloupdate]>
```

For whatever reason, if deployment status for one or more appliance need to be reset, following command can be executed

```
delete from siloupdate.appliance_deployment where mid=<mid> and patch_file_id=<patch_file_id>
```

`mid => Can be retrieved from System -> Settings -> Appliances Page`
`patch_file_id => Can be retrieved from master_logs.system_patches table or hover the mouse over Stage or Deploy Icon in System -> Tools -> Updates Page and URL link at browser's bottom left status bar will have this info`

Along with deployment status, If deployment logs also need to be cleared, following command can be executed

```
delete from siloupdate.appliance_deployment where mid=<mid> and patch_file_id=
```

`mid => Can be retrieved from System -> Settings -> Appliances Page`
`deploy_id => Can be retrieved from siloupdate.deployment_schedule`

What do I do when deployment timed out for one or more appliance?

- Run `monitor_deploy` and check if deployment is still running on that appliance

- If it is actively running give it some more time for it to complete(process tree updating actively). Do not trigger another deployment attempt until the process is complete.
- If it is completed, version update process will kick in (runs every 10 minutes) and update the UI with this appliance information
- If deployment process gets stuck with pigz(refer [System Update (sysup) - Known Issues & Troubleshooting#Known Issues]) issue, follow the work around
- If deployment process gets stuck on the script `91_docker_maintenance_reload_and_restart_containers.sh` there is likely a problem related to docker, run `systemctl status docker` and bring to the attention of SysUppers if more assistance is needed

What do I do when deployment failed due to yum upgrade error?

- Check `/data/logs/patcher.log` and look for possible reasons

> **Filter logs to a specific mid**
>
> `grep "mid=<mid>," /data/logs/patcher.log` will filter out logs specific to a given mid

- Check if there was a previous deployment attempt for this appliance which timed-out. This has happened a lot of times that a user re-deploys on a time-out. What happens during this time-sensitive period, is the fact the even after a time-out, the deployment process will continue running until is succeeds for fails. In case of a success, it'll remove the staged RPM packages and repository metadata (on CU and MCs). A subsequent attempt to deploy will be queued until the proc_mgr is unpaused. Once unpaused, the deployment process will fail simply because the repository metadata are not available. If this happens, we need to [reset the staging]and deploying state of the affected appliances, and then proceed with staging followed with deployment. This issue has been fixed in 10.1.0 release. Post 10.1.0, a subsequent deployment attempt after a timeout will detect whether there was a successful deployment of the same patch, and return immediately to the user with a success status.

Package conflicts

Package conflicts are caused when an installed RPM package conflicts with another package marked for upgrade. This is caused due to customers installing extraneous package(s) not provided by ScienceLogic on SL1 appliances. Customers should refrain from installing custom packages on SL1 appliances as they might conflict with the System Update, and it voids the warranty (see [https://support.sciencelogic.com/s/article/4017] for details).

- Look for conflicting packages in the log. In the example below, `shim-x64` package conflicts with the `kernel` package:

```
"2021-01-17 17:59:55,796","DEBUG","218180","PatchDeploy","patch_deploy","201","sid=4, mid=1,     Error: kernel conflicts with shim-x64-15-2.0.3.el7.x86_64"
"2021-01-17 17:59:55,796","DEBUG","218180","PatchDeploy","patch_deploy","201","sid=4, mid=1,     chmod: cannot access â€˜/var/lib/em7/.deployed_versionsâ€™: No such file or directory"
"2021-01-17 17:59:55,796","DEBUG","218180","PatchDeploy","patch_deploy","201","sid=4, mid=1,     ---> Package python34-devel.x86_64 0:3.4.10-4.el7 will be erased"
"2021-01-17 17:59:55,796","DEBUG","218180","PatchDeploy","patch_deploy","201","sid=4, mid=1,     ---> Package python34-libs.x86_64 0:3.4.10-4.el7 will be erased"
"2021-01-17 17:59:55,796","DEBUG","218180","PatchDeploy","patch_deploy","201","sid=4, mid=1,     ---> Package zeromq.x86_64 0:4.1.4-6.el7 will be erased"
"2021-01-17 17:59:55,796","DEBUG","218180","PatchDeploy","patch_deploy","201","sid=4, mid=1,     --> Processing Conflict: kernel-3.10.0-1160.el7.x86_64 conflicts shim-x64 <= 15-2.0.3.el7"
"2021-01-17 17:59:55,796","DEBUG","218180","PatchDeploy","patch_deploy","201","sid=4, mid=1,     --> Finished Dependency Resolution"
"2021-01-17 17:59:55,796","DEBUG","218180","PatchDeploy","patch_deploy","201","sid=4, mid=1,      You could try using --skip-broken to work around the problem"
"2021-01-17 17:59:55,796","DEBUG","218180","PatchDeploy","patch_deploy","201","sid=4, mid=1,      You could try running: rpm -Va --nofiles --nodigest"
"2021-01-17 17:59:55,796","DEBUG","218180","PatchDeploy","patch_deploy","201","sid=4, mid=1,     Yum upgrade failed with 1"
```

- Check yum history if there are any non-sl1 packages installed

> In the below example, non-sl1 packages were installed as part of yum transactions #2 and #6 and those extraneous packages are listed as part of next snippet.
> #3 is patch deploy and #4 & #5 are part of mariadb server package upgrade
> **Note:** This captures non-sl1 packages installed only via yum whereas the same could have been installed via rpm

From 11.1.0+

non-sl1 packages installed via yum

```
[root@dw-dr-db1-10-2-12-82 ~]# history_file=$(ls -1t /var/lib/yum/history/history-*.sqlite|head -1);sqlite3 $history_file "select tid, cmdline from trans_cmdline"
2|--nogpg -y install open-vm-tools
3|--assumeyes --nogpgcheck --disablerepo=* --enablerepo=em7,em7-os upgrade
4|--assumeyes --nogpgcheck --disablerepo=* swap -- remove galera -- install /tmp/umariadb-0I0-032/galera-4-26.4.6-1.el7.centos.x86_64.rpm /tmp/umariadb-0I0-032/socat-1.7.3.2-2.el7.x86_64.rpm
5|--assumeyes --nogpgcheck --disablerepo=* install /tmp/umariadb-0I0-032/MariaDB-server-10.4.17-1.el7.centos.x86_64.rpm
6|install telnet
[root@dw-dr-db1-10-2-12-82 ~]#
[root@dw-dr-db1-10-2-12-82 ~]#
[root@dw-dr-db1-10-2-12-82 ~]# repo_list=$({ yum list installed|awk '$3 ~ /^ *@/ {print $3}'  | sort  | uniq | grep -ivE '@anaconda|@em7*|@/galera-4|@/MariaDB-server|@/socat' ;} 2>/dev/null ); for repo in $(echo $repo_list) ;do yum list installed| grep $repo ; done
fuse.x86_64                             2.9.4-1.0.8.el7            @ol7_latest
fuse-libs.x86_64                        2.9.4-1.0.8.el7            @ol7_latest
libmspack.x86_64                        0.5-0.8.alpha.el7          @ol7_latest
open-vm-tools.x86_64                    11.0.5-3.0.1.el7_9.1       @ol7_latest
telnet.x86_64                           1:0.17-66.el7              @ol7_latest
xmlsec1.x86_64                          1.2.20-7.el7_4             @ol7_latest
xmlsec1-openssl.x86_64                  1.2.20-7.el7_4             @ol7_latest
[root@dw-dr-db1-10-2-12-82 ~]#
```

For older releases

packages that are listed under **<unset>** were installed using *rpm* and the rest of the packages were installed using *yum*. Run *yum history* and *yum history info <id>* for more information on these packages like install time, list of other packages etc

non-sl1 packages installed via yum

```
[root@dw-sm-db1-patch-10-2-12-113 ~]# sysuptb check-extra-pkgs
<unset>:
    htop-2.2.0-3.el7.x86_64
ol7_latest:
    libmspack-0.5-0.8.alpha.el7.x86_64
    open-vm-tools-11.0.5-3.0.1.el7_9.3.x86_64
    xmlsec1-openssl-1.2.20-7.el7_4.x86_64
    fuse-2.9.4-1.0.9.el7.x86_64
    fuse-libs-2.9.4-1.0.9.el7.x86_64
    xmlsec1-1.2.20-7.el7_4.x86_64
Found 7 extraneous packages installed on this appliance
[root@dw-sm-db1-patch-10-2-12-113 ~]#
```

There isn't any simple and straightforward steps to resolve package conflicts and steps would vary case by case depending on the kind of package installed and the dependency tree it brought in.
Also expectation is customers are NOT supposed to treat SL1 appliances as just another linux box and installing non-SL1 packages would break patcher.
For disposable appliances like Collectors and APs, the recommendation is to reinstall the appliance afresh.
EM-39134: Prevent users from innocently doing yum or rpm commands READY would address making yum package installations/updates more difficult and scary

- Verify that the package is a part of SL1 ISO or patch bundle
- Check if it's safe to remove the conflicting extraneous package. Note that on SL1 appliances, Yum is configured to find dependencies for the package being removed and remove those dependent packages as well if they are not required by any other packages installed on the appliance. The test here is to ascertain that removing the conflicting package does not remove other packages that would be required on the appliance

```
sudo yum --disablerepo=* --assumeno remove <conflicting_package>
```

In the example below, we are ***testing*** the removal of an extraneous package `openldap-clients` :

```
sudo yum --disablerepo=* --assumeno remove openldap-clients
Loaded plugins: silo-update
Resolving Dependencies
--> Running transaction check
---> Package openldap-clients.x86_64 0:2.4.44-22.el7 will be erased
--> Finished Dependency Resolution
--> Finding unneeded leftover dependencies
Found and removing 0 unneeded dependencies

Dependencies Resolved

================================================================================================================================================================================================================================================================================================================================
 Package                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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Size
================================================================================================================================================================================================================================================================================================================================
Removing:
 openldap-clients                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                x86_64
2.4.44-22.el7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                @ol7_latest
570 k

Transaction Summary
================================================================================================================================================================================================================================================================================================================================
Remove  1 Package

Installed size: 570 k
Exiting on user command
Your transaction was saved, rerun it with:
 yum load-transaction /tmp/yum_save_tx.2021-03-02.20-02.mH9Kxl.yumtx
```

As you can see there are no SL1-specific/dependent packages being removed.

- Remove the conflicting package ONLY if any SL1-specific packages were not removed in the previous attempt

```
sudo yum --disablerepo=* --assumeyes remove <conflicting_package>
```

In the example below, the extraneous package `openldap-clients` is actually being removed

```
sudo yum --disablerepo=* --assumeyes remove openldap-clients
Loaded plugins: silo-update
Resolving Dependencies
--> Running transaction check
---> Package openldap-clients.x86_64 0:2.4.44-22.el7 will be erased
--> Finished Dependency Resolution
--> Finding unneeded leftover dependencies
Found and removing 0 unneeded dependencies

Dependencies Resolved

================================================================================================================================================================================================================================================================================================================================
 Package                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 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Size
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Removing:
 openldap-clients                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                x86_64
2.4.44-22.el7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                @ol7_latest
570 k

Transaction Summary
================================================================================================================================================================================================================================================================================================================================
Remove  1 Package

Installed size: 570 k
Downloading packages:
```

```
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Erasing    : openldap-clients-2.4.44-22.el7.x86_64 1/1
  Verifying  : openldap-clients-2.4.44-22.el7.x86_64 1/1

Removed:
  openldap-clients.x86_64 0:2.4.44-22.el7

Complete!
```

Duplicate packages

- Check if there are duplicate packages:

```
sudo package-cleanup --dupes
```

- Remove duplicate packages if any

```
sudo package-cleanup --cleandupes --removenewestdupes
```

Look for any errors

What do I do when deployment failed due to post update error?

> New log file for post udpate
>
> Starting with 8.12, we introduced logging for post_update scripts which will help identify how long each of multiple postupdate scripts ran and was it success or failed

> There have been cases where post-update logs might not be complete, and this usually means that the deployment script was interrupted/killed before completion. Make sure you look for any custom jobs being run that might kill processes.

On that appliance where post update failed, check `/data/logs/post_update.log` which will list the last script it executed and which got failed. Depending the script which failed and the reason that caused the failure, appropriate action need to be taken.

Once the post update failure is addressed, re-run deployment for that appliance either from UI or via pcli (pcli can trigger patching operations for specific appliances unlike UI)

- To proceed with the deployment, follow these steps:
  - Reset the staging status of the affected appliance
  - Re-Stage the affected appliance
  - Re-Deploy the affected appliance. The deployment job should complete faster as the RPM upgrades have already completed. It will just run the post-update until completion or failure and other maintenance tasks.

How do I run postupdate script manually?

> In an ideal scenario, there should be no reason to run postupdate script manually as it gets triggered as part of deployment. Even in case of a failed deployment, re-running deployment from UI or `pcli` should invoke the postupdate script before marking the appliance as deployment complete.
>
> So its strongly advised to figure out the reason/failure that led to this situation before running postupdate scripts manually. When in doubt please check with #team-sysuppers before proceeding

On that appliance where post update need to be run manually

- If target appliance is a CDB and part of HA/DR, **enable** maintenance for the DRBD cluster

```
coro_config -> select apporpriate option to enable cluster maintenance
```

- Pause the proc_mgr/em7

```
sudo touch /tmp/.proc_mgr_pause
```

- Run postupdate.sh script

```
sudo /opt/em7/share/scripts/em7_postupdate.sh
```

- Unpause proc_mgr

```
sudo rm -f /tmp/.proc_mgr_pause
```

- Restart services

```
sudo systemctl daemon-reload
sudo systemctl restart php-fpm em7
```

- If target appliance is a CDB and part of HA/DR, **disable** maintenance for the DRBD cluster

```
coro_config -> select apporpriate option to disable cluster maintenance
```

What do I do when deployment failed due to crm error?

It simply means one of the following commands didn't work as expected.
There's not much that can be done except re-trying deployment or trying to execute one of the following commands manually and check what's going on with `crm` cluster

```
crm status
crm node maintenance
crm node ready
```


```
crm status
crm node maintenance
crm node ready
```


Version Update

Why deployed version is NOT updated on appliances page even though deployment was successful for that appliance?

How it works:

CDB will trigger deployment to remote appliance and o

- Checif patcher.log has version update errors for the given mid
- Check if the appliance in question is reachable via MySQL
  - silo_mysql -h <CU_IP> -P 7707 -u root -p<password>

```
[root@dw-sm-db1-patch-10-2-12-113 logs]# rpm -qa em7-patch-hook --dbpath /var/lib/em7/update/patch_hook/.rpmdb/
em7-patch-hook-8.13.0-428.smailappan_EM_26803_stop_building_el5_throwaway.x86_64
[root@dw-sm-db1-patch-10-2-12-113 logs]# rpm -q siloupdate
```

Others

How do I know what version of patch-hook/siloupdate (RPMs) is currently installed?

```
[root@dw-sm-db1-patch-10-2-12-113 logs]# rpm -qa em7-patch-hook --dbpath /var/lib/em7/update/patch_hook/.rpmdb/
em7-patch-hook-8.13.0-428.smailappan_EM_26803_stop_building_el5_throwaway.x86_64
[root@dw-sm-db1-patch-10-2-12-113 logs]# rpm -q siloupdate
siloupdate-0.1-201.x86_64
[root@dw-sm-db1-patch-10-2-12-113 logs]#
```

How do I fix corrupt RPM database?

Make sure there are no active RPM or yum processes(ps -eaf| grep -e 'yum|rpm') before continuing to the next steps:

```
mkdir -p /tmp/rpm.bak
cp /var/lib/rpm/* /tmp/rpm.bak
rm -f /var/lib/rpm/__db*
rm -f /var/lib/rpm/.*.lock
rpm --rebuilddb -vv
rpm -q kernel
```

If you are able to query the version of kernel RPM packages, you can safely remove the backup:

```
rm -Rf /tmp/rpm.bak
```

If issue still persists, check and clear the queued ha_status and system_status process as outlined here